

Abdelwahab Mahmoud

Senior Software Engineer · Backend & Platform

Cairo, Egypt

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Summary

Senior backend engineer with 8+ years building and owning production Ruby on Rails services used by real customers. Over the past two years, I've owned the multi-provider LLM engine, AWS infrastructure on ECS Fargate, and credit metering for a metered SaaS translation product, handling provider failover, retries, tracing, and evaluation to keep it reliable. I work backend-first across the full stack: highly scalable, performant Rails and Node.js services on AWS, asynchronous queueing and streaming between services, React frontends when needed, Datadog and OpenTelemetry for monitoring and observability, and data-intensive paths on Redis, MySQL, and PostgreSQL. I ship features end-to-end; when production breaks, I find the root cause and fix it, then harden the path so the failure does not repeat.

Work Experience

OnTheGoSystems

Remote (Hong Kong)

Senior Software Engineer

Dec. 2024 - Present

- Built the Rails translation engine across Claude on Bedrock, OpenAI and Gemini: provider fallback chains, JSON repair, Langfuse tracing, and rate-limit retries for the errors responsible for 65% of review failures.
- Shipped an agent-driven triage platform for production exceptions, solo in 12 weeks. Its pass/fail call lives in code, not the prompt, and caught three false passes when replayed on past incidents.
- Built the production tooling AI agents run behind: read-only data access with a rollback wrapper, an audit log, and a human-only write gate. Then the investigator and critic agents that adjudicate support claims.
- Migrated the product from a single EC2 host onto ECS Fargate, then owned that platform and its Terraform for 16 months, running 90% on Spot.
- Replaced ECS autoscaling with a queue-depth autoscaler on Lambda that drains workers rather than stopping them, because scale-in kills tasks abruptly and these jobs run from 30 seconds to 2 hours.
- Traced Aurora MySQL at 100% CPU to a status broadcast whose query plan examined 7.33M rows per call; dropped the join and throttled it from 1,013 calls to about 100.
- Owned credits and billing correctness for a metered LLM product, then designed one authorization architecture to replace its scattered credit checks—4 written ADRs, first shipping step merged.
- Wrote the ISO 27001 audit checklist and its risk profiles, then built the disaster recovery and data loss prevention programs to a documented 30-minute RTO and 15-minute RPO.

Dailymealz

Riyadh, Saudi Arabia

Senior Software Engineer

Dec. 2021 - Dec. 2024

- Built the real-time order service behind a 100,000 daily-user delivery product, on Node.js and Socket.IO, pushing kitchen preparation, driver pickup and delivery status to mobile and web clients.
- Decomposed order state out of the Laravel monolith into a standalone Node.js service in an event-driven, Service-Oriented Architecture (SOA), exchanging state changes over SQS with a REST API contract for subscribed clients.
- Served hot order state from Redis as the service's read source with MySQL behind it, keeping status fan-out off the transactional path.
- Migrated the service from JavaScript to TypeScript.
- Led a team of 5 building a background driver location check-in system: planning, reviews and pairing across the driver app, cutting manual check-in overhead from the routing flow.
- Instrumented order, driver and kitchen flows with Datadog monitors and OpenTelemetry, cutting Mean Time To Resolve 30%.

Nformacy

Cairo, Egypt

Full-Stack Developer

Oct. 2020 - Dec. 2021

- Led features from business requirements to deployment, backend and frontend.
- Moved the web apps to a SaaS model and built mentor booking with Zoom, ClickUp and calendar integrations.
- Wrote BDD tests in Cypress and managed cloud infrastructure and deployment pipelines.

DevSquads (formerly Fikr Labs)

Cairo, Egypt

Software Engineer

Jan. 2018 - Oct. 2020

- Built software in self-organizing Extreme Programming teams, using TDD, thin vertical slices and test-covered refactoring of legacy code.
- Developed across React Native for mobile, Ruby on Rails and Java Spring for backend, and React for frontend.
- Mentored teams adopting XP and Agile practices: pairing sessions, code review discipline, iterative planning.

Skills

Programming Languages	Ruby, TypeScript, JavaScript, Java (Spring), Python, SQL, PHP
Backend Development	Ruby on Rails, Node.js, Express, Socket.IO, REST API design, service-oriented architecture, background workers, FastAPI
Frontend Development	React, Next.js, Redux, Hotwire (Turbo & Stimulus), React Native, Tailwind CSS
AI & LLM	AWS Bedrock, OpenAI, Gemini, agentic workflows, Claude Agent SDK, MCP, Langfuse, evaluation harnesses
Cloud & Infrastructure	AWS (ECS Fargate, Lambda, SQS, Step Functions, CloudWatch, S3, IAM), Terraform, Docker, GitLab CI
Observability	Datadog, OpenTelemetry, AWS CloudWatch, Langfuse tracing
Databases	PostgreSQL, MySQL & Aurora, Redis, DynamoDB
Testing	TDD (Test-Driven Development), RSpec, Capybara, Playwright, Cypress, Jest, VCR
Spoken Languages	English, Arabic (native), German (beginner)

Education

Faculty of Science Cairo University

Bachelor of Science in Geophysics

Cairo, Egypt
Sep. 2012 - Jun. 2016